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## **An Efficient Chelator - MGDA to Remove Calcium Sulfate Scales in Oil Production Systems**

J. Hou and T. Huang, Beijing Research Center, Aramco Asia, Beijing, China; T. Chen, Saudi Aramco, Dhahran, Saudi Arabia; C. Da, Beijing Research Center, Aramco Asia, Beijing, China

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### **Abstract**

Scales can accumulate and obstruct flow pathways in oil production systems, causing production rates reduction, energy consumption increasing and operational disturbances. The mitigation of scale deposition is costly but pivotal for maintaining high-performing production processes. Calcium sulfate, an acid-insoluble scale, requires efficient chelator to be removed. Herein, this paper demonstrates the effectiveness of trisodium dicarboxymethyl alaninate (MGDA) to dissolve  $\text{CaSO}_4$  at various temperatures, presenting the scenarios of downhole and flow lines in oil production system.

$\text{CaSO}_4$  scale removal tests were conducted by immersing  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  crystals (10 mm \* 2 mm) in solutions for a certain time at 50°C (representing surface flow line conditions) and 100°C (presenting wellbore conditions) respectively. The original and residual weight of crystals were measured to calculate the removal efficiency. Factors like MGDA concentration, scale/solution ratio and dissolving rate in a time scale were also evaluated to present the chelator performance and dissolving mechanism. Besides, whether the addition of  $\text{K}_2\text{CO}_3$  has positive effects on  $\text{CaSO}_4$  removal is investigated.

MGDA is selected to be investigated since it presented a higher removal efficiency comparing with Glutamic acid diacetate- $\text{Na}_4$  (GLDA) and penta-sodium diethylene-triaminepentaacetic acid (DTPA) in the lab screening. Results showed the scale removal efficiency using 8% MGDA at 100 °C were 83% and 42% at scale/solution ratios of 1/10 and 2/10 respectively after 24h treatment. At 50 °C, the scale efficiency decreased to 30% at scale/chemical ratio 1 /10 ratio, but the value is over 60% at 0.5/10 ratio. It is noted that increasing MGDA concentration will not obviously improve the crystal dissolving efficiency. Thus, it is recommended a higher solution volume at low temperature conditions. Besides, the time scale results presented the dissolving efficiency quickly increased in the first hour and then slowly increased with time. Furthermore, addition of potassium carbonate and subsequent acid washing of the residual solids will promote the conversion of  $\text{CaSO}_4$  to  $\text{CaCO}_3$  for acid removal, which slightly improved the removal efficiency at low addition concentrations. High concentration addition of potassium carbonate will deteriorate the scale dissolving by MGDA. This work demonstrates MGDA as an effective chelator in  $\text{CaSO}_4$  removing at various temperatures, which is a potential selection for future oilfield operations.

## Introduction

Sulfate scales are formed when two incompatible waters mix, with one containing a high concentration of sulfate ions and the other rich in calcium, barium, and/or strontium ions (Vavouraki et al., 2008). These scales can develop in oil and gas wellbores, as well as in surface flow lines, causing formation damage, equipment issues, and blockages in pipelines and surface facilities. This, in turn, leads to a reduction in oil and gas production. (Bader, 2007; Van Driessche et al., 2019) In this condition, scale mitigation methods are sometimes necessary to guarantee the produced fluid flow. (Chen et al., 2024; Chen et al., 2023; Hou et al., 2022) While the methods can be chemical or physical, chemical methods are considered because the relative low cost and easy application of chemicals, and suitable for various scale treatments, such as  $\text{CaCO}_3$ ,  $\text{CaSO}_4$ ,  $\text{BaSO}_4$  etc.

Among the scales found in oil production, calcium sulfate ( $\text{CaSO}_4$ ) is one of the most common sulfate scales in carbonate reservoirs (Mohammed et al., 2024). However, the removal of this scale is costly, since it cannot dissolve in acid. A number of chemicals have been identified for the removal of  $\text{CaSO}_4$ , primarily consisting of chelating agents and other auxiliary chemicals (Alissa et al., 2022; Alqahtani et al., 2023; Clemmit et al., 1985). Commonly used chelators include ethylenediaminetetraacetic acid (EDTA), hydroxyethyl ethylene diamine triacetic acid (HEDTA), glutamic acid diacetic acid (GLDA), diethylenetriamine pentaacetic acid (DTPA) and methylglycinediacetic acid (MGDA) (Sokhanvarian et al., 2012). The main chemical routes have been identified for dissolving  $\text{CaSO}_4$  include 1) Chelating agents enhance the solubility of  $\text{CaSO}_4$  by complexing the  $\text{Ca}^{2+}$  ions from the scales, tipping the balance in favor of dissolution; 2)  $\text{CaSO}_4$  can be converted to  $\text{CaCO}_3$  in the presence of  $\text{CO}_3^{2-}$  ions since  $\text{CaCO}_3$  is more stable and has a lower precipitation coefficient  $K_{sp}$ ; and 3) dissolved calcium ions precipitate as  $\text{Ca(OH)}_2$  at high alkaline conditions and promote more calcium ions to dissolve in the solution. Accordingly, the removal is usually effective at high temperature high pH conditions.

Mixtures of EDTA with  $\text{K}_2\text{CO}_3$  or  $\text{Na}_2\text{CO}_3$  are among the most popular formulations. Murtaza et al. reported the use of potassium carbonate and tetrapotassium-EDTA at 200°F for single-step removal (Murtaza et al., 2022). Gamal introduced a new scale dissolver composed of acrylic acid and lactic acid at pH 9, which is non-corrosive and exhibits high scale dissolution performance for composite scales (Gamal et al., 2021).

The primary drawbacks of EDTA and similar chelating agents are: 1) their poor solubility in mild pH conditions, which requires very high concentrations of alkaline substances to achieve a high chelator concentration solution, leading to environmentally harmful alkaline wastes; and 2) their lack of biodegradability. For instance, ethylenediaminetriacetic acid (ED3A) is one of the decomposition products of EDTA, which can then cyclize to form a persistent pollutant 3-ketopiperazine-N,N-diacetate (3KP) (Yuan and VanBriesen, 2006).

Biodegradation is the primary method for removing residual chelating agents and metal complexes from the environment, thereby preventing toxicity to soil, water bodies, and human health (Diarra et al., 2021). Due to the health and environmental side effects of certain chelators, ongoing research aims to replace these chelators with biodegradable alternatives that are effective at mild pH conditions. The development and use of biodegradable and environmentally friendly chelating agents for scale removal, with consideration for wastewater treatment before disposal and the environmental requirements of offshore applications, represents a promising advancement.

Trisodium dicarboxymethyl alaninate or MGDA, which is a commonly used biodegradable chelator in industry. It has presented high  $\text{Ca}^{2+}$  and  $\text{Ba}^{2+}$  chelating and scale removal potential in literature. (Bretti et al., 2017a; Bretti et al., 2017b) To further show its ability in  $\text{CaSO}_4$  removal, in this paper, we compared the  $\text{CaSO}_4$  scale removal efficiency with GLDA and DTPA using gypsum crystals. Then the initial solid/solution volume ratio and dynamic dissolving at a time scale were investigated at both 50 °C and 100 °C. In

one word, this work shows the biodegradable chelating agent MGDA is a highly efficient  $\text{CaSO}_4$  remover for oil and gas applications.

## Experiment

### Chemicals

MGDA (40%), GLDA (40%) and DTPA (50%) were collected from local market and used without further purification.  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  are crystals with average size of  $2 \text{ mm} \times 2 \text{ mm} \times 10 \text{ mm}$  (Fig. 1).



Figure 1— $\text{CaSO}_4$  crystals used in the tests.

### Removal efficiency testing

In the removal efficiency tests, 2g  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  crystals were immersed in 10 mL chemical samples and incubated at 25 °C, 50 °C and 100 °C in a static state respectively. After 24h incubation, the upper solution is dropped and the residual solids are washed by water for three times. Then, the residual  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  are dried at 50 °C overnight. And the weights of the dried solids are measured with precision of  $\pm 0.01 \text{ g}$ . In the dosage optimization, the volume of chemical solution is kept as 10 mL, and the  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  mass is from 0.5 g to 4.0 g. In the dynamic scale removal test, the incubation time of chemical solution and  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  crystals is from 2h to 24h. For the evaluation of dispersant type chemicals, weight measurement method is not used since the residual mass increases compared to original mass. Visual observation of slurry is used to select the best chemical.

$$\text{Removal efficiency} = (\text{original mass} - \text{residual mass}) / \text{original mass} * 100\%$$

### $\text{CaSO}_4$ solubility test

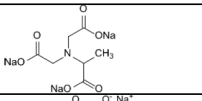
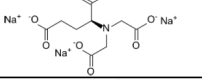
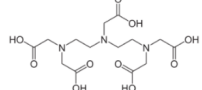
$\text{CaSO}_4$  powders (pure chemical) instead of crystals are weighted as 1 g and added into 8% MGDA solution. The mixtures are stirred at 500 rpm at room temperature or 50 °C for 24 hours. For 100 °C measurements, the mixtures are added to a penicillin bottle, sealed and put in the 100 °C oven for 24 hours. To make sure a complete dissolving, the bottle was shaken every 30 mins in the first 6 hours incubation. After the dissolution process, the residual solids are removed by filtration through a  $0.45 \mu\text{m}$  membrane. The  $\text{Ca}^{2+}$  concentration is determined by inductively coupled plasma mass spectrometry to calculate the mass of total dissolved  $\text{CaSO}_4$ .

## Results and discussions

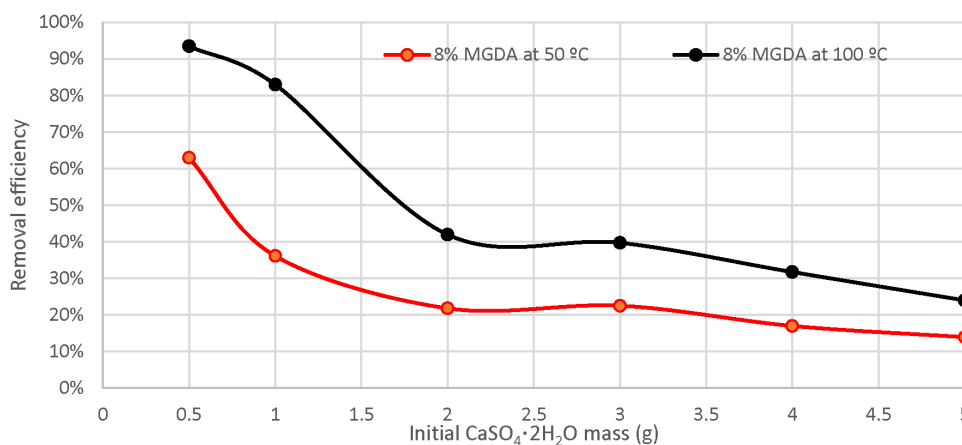
MGDA and GLDA are biodegradable chelating agents, while DTPA, a non-biodegradable chelator, is chosen for its stronger chelating performance compared to EDTA. The pH values of the dilute solutions of these chelating agents, all in their sodium salt states, are above 12, as shown in Table 1. To assess their  $\text{CaSO}_4$  scale dissolving abilities, the efficiency of MGDA, GLDA, and DTPA was measured at 25 °C, 50 °C, and 100 °C, corresponding to different operational conditions such as surface flow lines (25 °C or 50 °C)

and downhole environments (100 °C). At these temperatures, the stability of the chelating agents is not a primary factor for scale removal (Mahmoud et al., 2011). Table 1 indicates that removal efficiency increases with temperature for all chelating agents. At 25 °C, MGDA showed the highest efficiency at 22%, meaning only 22% of  $\text{CaSO}_4$  was removed. At 50 °C, the efficiency slightly increased for all agents. At 100 °C, the chelating agents' ability to dissolve  $\text{CaSO}_4$  ranked as  $\text{MGDA} > \text{GLDA} > \text{DTPA}$ .

**Table 1— $\text{CaSO}_4$  scale removing efficiency by using MGDA, GLDA and DTPA at 25 °C, 50 °C and 100 °C, respectively.**

| Chelator | Concentration w% | pH   | At 25 °C | At 50 °C | At 100 °C | Chemical structure                                                                  |
|----------|------------------|------|----------|----------|-----------|-------------------------------------------------------------------------------------|
| MGDA     | 8%               | 12.8 | 0.22     | 0.24     | 0.42      |  |
| GLDA     | 8%               | 12.4 | 0.20     | 0.23     | 0.39      |  |
| DTPA     | 10%              | 12.8 | 0.17     | 0.19     | 0.32      |  |

Next, we evaluated the solid/solution ratio on the  $\text{CaSO}_4$  scale removal efficiency at 50 °C and 100 °C respectively. MGDA is used in this experiment since it presented the highest performance (Table 1). With the constant volume of 8% MGDA at 10 mL, 0.5 g, 1 g, 2 g, 3 g, 4g and 5 g crystals were immersed in the solution. It shows MGDA performed an excellent  $\text{CaSO}_4$  removal efficiency in the solid/solution ratio of 0.5g/10mL and 1g/10mL at 100 °C, with the value of 93% and 83% respectively (Fig.2). Although the removed crystals increased with the  $\text{CaSO}_4$  initial weight when the solid/solution ratio is less than 3g/10mL, the resultant efficiency decreased quickly when the initial weight was reduced below 2 g. At 4g/10mL and 5g/10mL, the removed mass was similar to that at 3g/10mL. This indicates that when the solid/solution is lower than 3g/10mL, the kinetic became the dominating factor to the efficiency of  $\text{CaSO}_4$  removal by the chelating agent solutions. When the solid/solution is higher than 3g/10mL, of the removal efficiency reached the plateau for the MGDA solutions. At 50 °C, the curves showed the similar trend as at 100 °C.



**Figure 2—Removal efficiency of 8% MGDA at different initial  $\text{CaSO}_4$  weight at 50 °C and 100 °C. (Chemical volume is fixed at 10 mL).**

Next, we evaluated the effect of the solid/solution ratio on  $\text{CaSO}_4$  scale removal efficiency at 50 °C and 100 °C. MGDA was used in this experiment as it demonstrated the highest performance compared to GLDA (Table 1). With a constant volume of 8% MGDA at 10 mL, varying amounts of crystals (0.5 g, 1 g, 2 g, 3

g, 4 g, and 5 g) were immersed in the solution. The results showed that MGDA achieved excellent  $\text{CaSO}_4$  removal efficiency at a solid/solution ratio of 0.5 g/10 mL and 1 g/10 mL at 100 °C, with efficiencies of 93% and 83%, respectively (Fig. 2). Although the amount of removed crystals increased with the initial weight of  $\text{CaSO}_4$  when the solid/solution ratio was less than 3 g/10 mL, the efficiency decreased rapidly when the initial weight was reduced below 2 g. At 4 g/10 mL and 5 g/10 mL, the removed mass was similar to that at 3 g/10 mL, indicating no more crystals can be removed. This indicates that when the solid/solution ratio is lower than 3 g/10 mL, kinetics becomes the dominant factor in the efficiency of  $\text{CaSO}_4$  removal by the chelating agent solutions. At 50 °C, the trends were similar to those observed at 100 °C.

Fig.3 presents the solid/solution ratio on removal efficiency in 10 mL 20% MGDA solutions. The curves at 50 °C and 100 °C also shows the removal is efficient when solid/solution ratio is less than 3 g/10 mL. It is interesting that the increasing of MGDA concentration from 8% to 20% did not result to a higher efficiency. This may be because at high pH conditions, the formation of large amounts of  $\text{Ca(OH)}_2$  prevented the contacting and dissolving of Ca ions with MGDA. Lots of the MGDA molecules were in the sodium state.

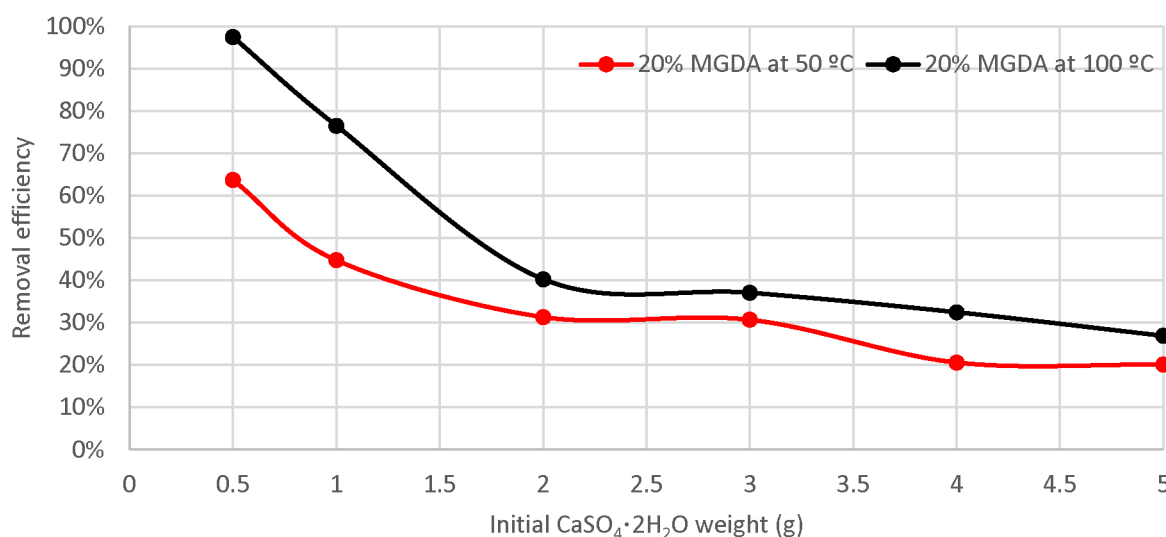


Figure 3—Removal efficiency of 20% MGDA at different initial  $\text{CaSO}_4$  weight at 50 °C and 100 °C. (Chemical volume is fixed at 10 mL).

Fig. 4 shows the  $\text{CaSO}_4$  solubility in 8% MGDA at different temperatures respectively. Since particles instead of crystals are used in the solubility test, it is likely the results are seldom affected by dynamic factors. It shows the value at 100 °C was only slightly higher than at 25 °C, 50 °C and 75 °C. This experiment presented that temperatures may have no obvious impact on the  $\text{CaSO}_4$  solubility in this range. Comparing with Fig.2, that the efficiency is obviously higher at 100 °C than at 50 °C, it is assumed that the removal of crystals is mainly determined by dynamic factors.

Besides, Fig.4 shows the solubility of  $\text{CaSO}_4$  in 8% MGDA at 100 °C is 50,651 mg/L, which means at most 0.678 g  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ . When at 2 g/ 10 mL and 3 g/ 10 mL at 100 °C cases, the removed  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  crystals were 0.84 g and 1.2 g, respectively, which are both higher than the solubility. This indicates that partially crystals are removed by forming  $\text{Ca(OH)}_2$  instead of dissolving with chelating agents.

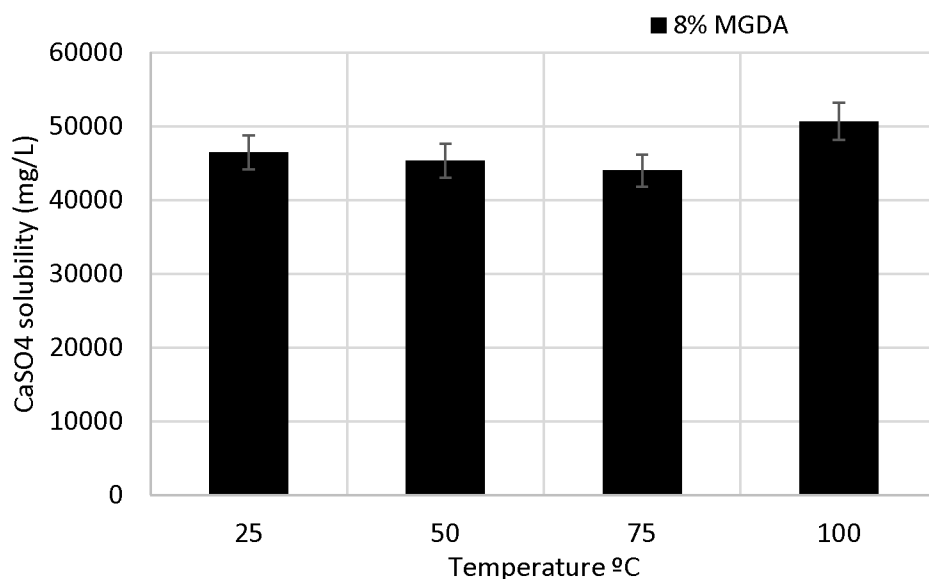


Figure 4—CaSO<sub>4</sub> solubility of 8% MGDA at 25 °C, 50 °C, 75 °C and 100 °C, respectively.

In summary, when using chelating agents under alkaline conditions, the small proportion of free Ca<sup>2+</sup> ions from dissolving calcium sulfate will either combine with the chelating agent to form a complex or react with OH<sup>-</sup> to form Ca(OH)<sub>2</sub>. (Shukla et al., 2008) This reaction further promotes the dissociation of CaSO<sub>4</sub>·2H<sub>2</sub>O into Ca<sup>2+</sup> and SO<sub>4</sub><sup>2-</sup>. Since the solubility of Ca(OH)<sub>2</sub> is much higher than that of CaSO<sub>4</sub>, with 1.65 g/L versus 0.24 g/L at 20 °C (Pallagi et al., 2012), the initially formed Ca(OH)<sub>2</sub> will not precipitate again.

Fig.5 presents the scale removal efficiency using MGDA at 50 °C and 100 °C in a time scale, respectively. 1 g crystals were used here. Both curves present a relatively high dissolving rate in the first 4 hours and then the rate decreased. And it is reasonable that the dissolving rate at 100 °C is higher than that at 50 °C. For 50 °C dissolving, no more scales are removed after 8 hours. The dissolved Ca<sup>2+</sup> concentration in solutions also showed a fast increase in the initial several hours and the increasing rate became slow. The decrease of dissolving rate could be explained by two possible reasons. One is kinetic factor that with the increase of Ca and decrease of MGDA concentrations, the reaction rate decreased. The other is the precipitates of Ca(OH)<sub>2</sub> on the surface of crystals, preventing the dissolving process of Ca and chelating effect.

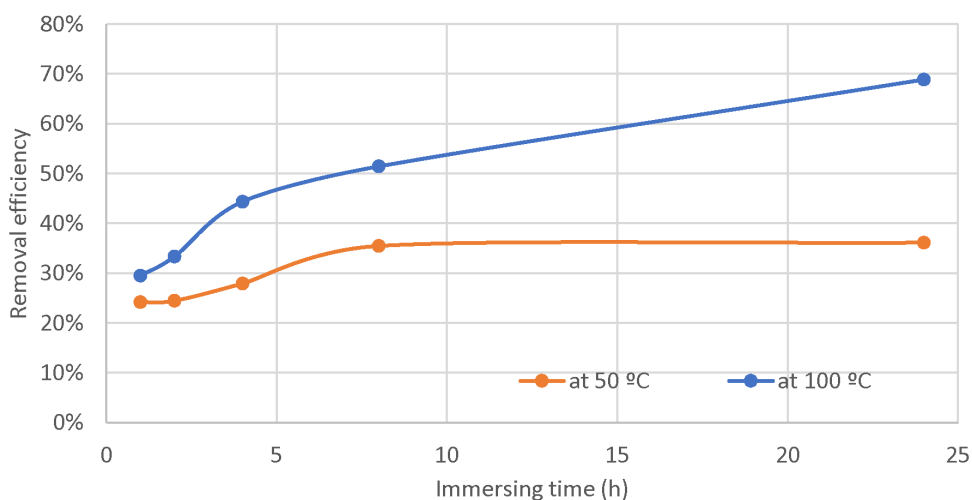
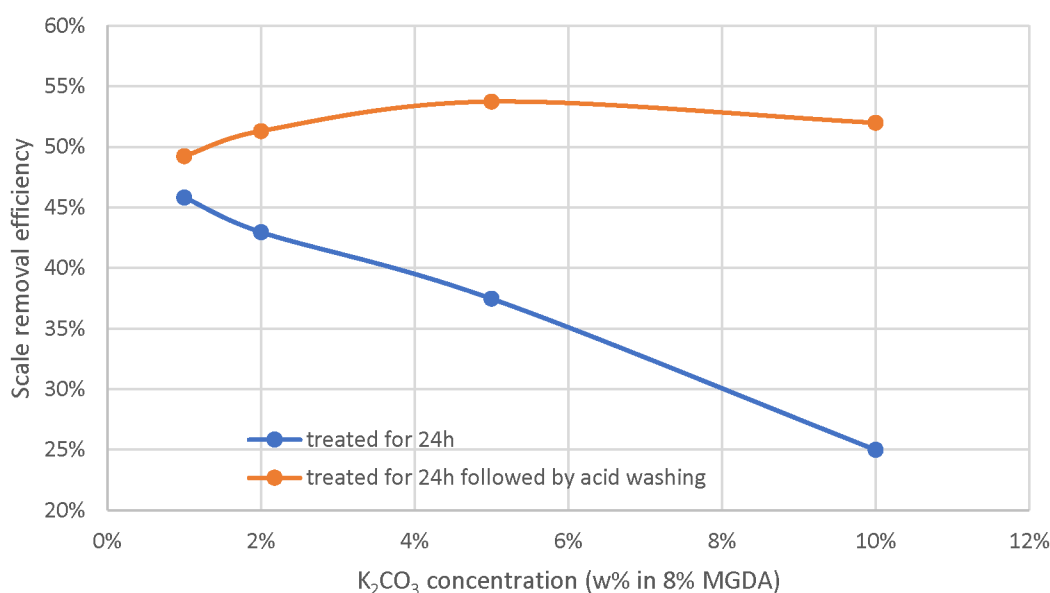


Figure 5—Removal efficiency of MGDA vs time at 50 °C and 100 °C. (CaSO<sub>4</sub> is 1 g in 10 mL chemical.)



Since the formation of  $\text{Ca}(\text{OH})_2$  improves scale removal, we further investigated whether the addition of carbonate ions could enhance scale removal efficiency by converting  $\text{CaSO}_4$  into  $\text{CaCO}_3$ . Fig.6 shows the conditions of adding different concentrations of potassium carbonate to MGDA, while keeping the MGDA concentration constant at 8% at 100 °C. When 1%  $\text{K}_2\text{CO}_3$  was added, the removal efficiency reached 46%, slightly higher than the 42% efficiency without  $\text{K}_2\text{CO}_3$ . After collecting, drying, and treating the residual solids with dilute acetic acid,  $\text{CaCO}_3$  dissolved, and the overall removal efficiency increased to 49%. However, with further addition of  $\text{K}_2\text{CO}_3$ , the removal efficiency decreased as more  $\text{CaCO}_3$  formed and remained as residual solids. Acid washing effectively removed these residual solids. The highest removal efficiency, 54%, was achieved with 5%  $\text{K}_2\text{CO}_3$  in MGDA. Further addition of  $\text{K}_2\text{CO}_3$  promoted  $\text{CaCO}_3$  formation initially but did not improve overall removal efficiency, possibly because the formed  $\text{CaCO}_3$  inhibited the reaction of  $\text{CaSO}_4$ . In summary, adding a small amount of carbonate salts to the chelating agent can slightly improve  $\text{CaSO}_4$  removal efficiency.



**Figure 6—Scale removal efficiency by adding different concentrations of  $\text{K}_2\text{CO}_3$  in MGDA. The test condition is 2 g crystals in 10 mL chemical for 24 hours at 100 °C.**

## Conclusions

With the worldwide increasing demand of biodegradable chelating agents as the alternatives to traditional non-degradable aminopolycarboxylates and phosphonates, this work provides biodegradable chelating agents, MGDA with potential to replace traditional chelators for  $\text{CaSO}_4$  removal in oil and gas industry. It concludes that

1. By investigating the performance of MGDA, GLDA and DTPA on  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  crystals dissolving respectively, removal efficiency turned out to be ranked as  $\text{MGDA} > \text{GLDA} > \text{DTPA} > \text{HEDP}$ .
2. The optimal solid/solution ratio is 1 g/ 10 mL or 2 g/ 10 mL MGDA. High concentration of MGDA, such as 20%, is found unable to improve the scale removal efficiency than 8%.
3. The removing mechanism under alkaline conditions regards on combination of Ca ions with the chelating agent to form a complex and the reaction of Ca ions with  $\text{OH}^-$  to form  $\text{Ca}(\text{OH})_2$ .

Overall, the applications of biodegradable chelating agents are essential to the oil and gas industry with a green future.

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